2023 Ch H1 Q13

Section: Nature's Chemistry

Topic: Esters

Question Summary

Two flasks, A and B, were placed in a water bath at 40°C.
Flask A contained ethanol and ethanoic acid.
Flask B contained ethyl ethanoate and water.
After several days, the contents of the flasks were analysed. Which results would be expected?

Worked Solution

Ethanol + ethanoic acid ⇌ ethyl ethanoate + water (esterification/hydrolysis equilibrium).

• Flask A starts with reactants: ethanol + ethanoic acid. At equilibrium, it will contain a mixture of all four species: ethanol, ethanoic acid, ethyl ethanoate, and water.

• Flask B starts with products: ethyl

ethanoate + water. At equilibrium, it will also contain all four species.
Therefore, after several days, both flasks will contain all four substances.

Final Answer

D — Flask A and flask B contain ethyl ethanoate, water, ethanol and ethanoic acid

Revision Tips

• Esterification is a reversible reaction.
• At equilibrium, both forward and reverse reactions occur at the same rate.
• Regardless of starting point (reactants or products), the equilibrium mixture contains all four species.